

MEMBER OF THE UNIT DEVLAB ECOSYSTEM

# UNIT PULSAR ESP32-C5

MPN: UE0134

Version 0.1.0



ESP32-C5 development board with dual-band Wi-Fi 6, Bluetooth LE, 802.15.4, environmental sensing, battery fuel gauge, microSD, QWIIC, and USB-C.

ESP32-C5HR8

8 MiB FLASH

8 MB PSRAM

WI-FI 6 · BLE · 802.15.4

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# Description

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The UNIT PULSAR ESP32-C5 is a multi-interface development board in the UNIT DevLab ecosystem, built around the Espressif ESP32-C5HR8 SoC with 8 MB of in-package PSRAM. The V1.2.3 design combines dual-band 2.4/5 GHz Wi-Fi 6, Bluetooth LE, and IEEE 802.15.4 connectivity with external QSPI flash, environmental and magnetic sensing, microSD, USB-C, Li-ion charging with fuel gauging, an addressable RGB indicator, QWIIC I2C, and an RF coaxial antenna connector.

## Applications

- Dual-band Wi-Fi 6, Bluetooth LE, Zigbee, and Thread prototyping
- Environmental monitoring and data logging
- Battery-powered IoT nodes with fuel gauging
- Magnetometer and proximity-based user interfaces
- microSD data logging
- Education and subsystem validation

## Hardware Features

- ESP32-C5HR8 SoC: 32-bit RISC-V up to 240 MHz, 384 KB HP SRAM, 8 MB PSRAM
- BY25Q64ES 64 Mbit (8 MiB) quad SPI NOR flash
- 2.4/5 GHz RF path through a diplexer to a coaxial antenna connector
- LPS22 pressure, FHT40 humidity/temperature, VCNL4040 proximity/light, and MMC5603NJ magnetometer sensors
- BQ24074 Li-ion charger with power path and MAX17048 fuel gauge
- SGM6029 1 A buck regulator for the 3.3 V rail
- microSD socket on SPI signals
- One WS2812 RGB LED plus power, charge, and user indicators
- USB-C, QWIIC, 6-pin SPI, battery, and Nano-style edge-pad connections

# 1 The Board

The UNIT PULSAR ESP32-C5 is a development board for evaluating the Espressif ESP32-C5HR8 and building connected applications that combine dual-band Wi-Fi 6, Bluetooth LE, and IEEE 802.15.4 radio with environmental sensing, removable storage, battery operation, and general-purpose I/O. Schematic revision V1.2.3 places the SoC, flash, RF diplexer, power-management devices, sensors, indicators, and user controls on a Nano-style board with two parallel edge-pad rows.

## 1.1 Accessories

The following optional accessories are recommended for use with the UNIT PULSAR ESP32-C5. Select accessories according to the interfaces and features required by the application.

| Accessory                                    | Purpose                                           | Selection notes                                                                                           |
|----------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Dual-band antenna with matching coaxial plug | Wi-Fi, Bluetooth LE, and 802.15.4 radio operation | Must cover 2.4 GHz and 4.9–5.95 GHz and mate with the J2 RF1-2626 connector; no onboard antenna is fitted |
| QWIIC cable                                  | External I2C expansion                            | 4-pin, 1 mm pitch cable for connecting compatible 3.3 V QWIIC devices                                     |
| USB-C data cable                             | Power, programming, and USB serial                | Must support data; a charge-only cable cannot upload firmware                                             |
| MicroSD card                                 | Removable storage and data logging                | Class 10 microSD card; FAT32 formatting is recommended                                                    |
| LiPo 3.7 V 650 mAh Battery                   | Battery-powered operation                         | Single-cell LiPo battery; verify polarity and connector orientation before connection                     |

Verify connector type, orientation, polarity, voltage domain, and pinout before connecting accessories.

## 1.2 Board Identification

| Item                          | Value                                      |
|-------------------------------|--------------------------------------------|
| Product                       | UNIT PULSAR ESP32-C5                       |
| SKU                           | UE0134                                     |
| Product family                | UNIT DevLab ecosystem                      |
| Product type                  | Multi-interface ESP32-C5 development board |
| Main component                | Espressif ESP32-C5HR8, QFN-48 (6 × 6 mm)   |
| Hardware revision             | V1.2.3 (schematic)                         |
| Product Reference             | Version 0.1.0                              |
| Primary programming interface | USB-C / ESP32-C5 USB Serial/JTAG           |
| Debug interface               | USB Serial/JTAG through USB-C              |

Hardware revision and documentation revision are identified independently.

## 1.3 Main Assemblies

| RefDes          | Component                | Function                                                |
|-----------------|--------------------------|---------------------------------------------------------|
| IC1             | ESP32-C5HR8              | Main SoC with 8 MB in-package PSRAM and 2.4/5 GHz radio |
| IC3             | BY25Q64ESHIG             | 64 Mbit (8 MiB) external quad SPI flash                 |
| IC2             | BQ24074RGTR              | Single-cell Li-Ion charger with power path              |
| U2              | SGM6029CYG               | 1 A buck regulator for the 3.3 V rail                   |
| IC4             | MAX17048G+T10            | Battery fuel gauge                                      |
| A2              | LPS22HBTR                | Barometric pressure sensor                              |
| U\$36           | FHT40-DD-TR              | Humidity and temperature sensor                         |
| U3              | VCNL4040M3OE             | Proximity and ambient-light sensor                      |
| U4              | MMC5603NJ                | Three-axis magnetometer                                 |
| U1              | LFD182G45DCHD277         | 2.4 GHz / 5 GHz RF diplexer                             |
| J2              | RF1-2626-125CE-CT        | RF coaxial antenna connector                            |
| XTAL1           | SX0B48.000F0810F30       | 48 MHz main crystal                                     |
| MICRO_SD-HOLDER | 47309-2651               | Removable microSD storage                               |
| LED1            | WS2812 1010              | Addressable RGB indicator                               |
| J1              | HCZZ0032-4               | Four-position QWIIC-style I2C connector                 |
| J20             | SM06B-SRSS-TB            | Six-position, 1 mm SPI expansion connector              |
| JP1, J3         | PH 2.0 mm 2P, 1.25 mm 2P | Parallel battery connections                            |
| S1, S2          | 1TS026A-1600-0553-CT     | BOOT and RESET push buttons                             |

## 1.4 Board Views

Top-side and bottom-side board photographs for revision V1.2.3 are not yet available. Refer to the schematic in Section 9.1 for component designators.

## 1.5 Handling

Handle the board using normal ESD precautions. Avoid touching the RF connector, sensor apertures, connector contacts, or exposed pads. Remove power before connecting or removing the battery. Connect an antenna to J2 before enabling radio transmission. Keep conductive objects away from the battery and microSD areas when the board is energized.

## 2 Ratings

This chapter defines the recommended operating conditions and electrical characteristics of the UNIT PULSAR ESP32-C5 hardware V1.2.3. Ratings apply to the complete board unless otherwise stated.

### 2.1 Recommended Operating Conditions

The UNIT PULSAR ESP32-C5 can be powered from USB-C, VIN, or a single-cell LiPo battery. The BQ24074 power-path charger feeds the onboard SGM6029 buck regulator, which supplies the 3.3 V logic domain used by the ESP32-C5HR8 and onboard peripherals.

| Parameter                         | Recommended operating condition | Notes                                                                                    |
|-----------------------------------|---------------------------------|------------------------------------------------------------------------------------------|
| USB-C input                       | 5 V nominal                     | Primary power and programming interface                                                  |
| VIN input                         | 5 V nominal                     | Joins USB VBUS through Schottky diode D4; board-level range not characterized            |
| Battery input                     | 3.7 V nominal, 4.2 V maximum    | Single-cell LiPo battery                                                                 |
| Regulated logic rail              | 3.3 V nominal                   | Generated by the onboard SGM6029 buck regulator                                          |
| 3.3 V regulator output capability | Up to 1 A                       | SGM6029 component rating; shared by the SoC, peripherals, LEDs, and external 3.3 V loads |
| Battery charge current            | 250 mA nominal                  | Set by R13 (3.57 k $\Omega$ ) on the BQ24074 ISET pin                                    |
| Input current limit               | $\approx 1.4$ A                 | Set by R12 (1.1 k $\Omega$ ) on the BQ24074 ILIM pin                                     |
| ESP32-C5 CPU clock                | Up to 240 MHz                   | Firmware configurable                                                                    |
| Main crystal                      | 48 MHz                          | XTAL1                                                                                    |
| Internal SRAM                     | 384 KB HP SRAM, 16 KB LP SRAM   | ESP32-C5 internal memory                                                                 |
| External flash                    | 64 Mbit / 8 MiB                 | BY25Q64ESHIG                                                                             |
| PSRAM                             | 8 MB                            | In-package, ESP32-C5HR8                                                                  |
| Logic level                       | 3.3 V                           | Applies to onboard digital interfaces and GPIO                                           |

### 2.2 Power Domains and Power Distribution

The UNIT PULSAR ESP32-C5 uses separate power domains for external supply, battery operation, system power, and regulated 3.3 V logic.

The main power domains are:

- **USB-C / VBUS:** provides the nominal 5 V USB supply used for board power, programming, and battery charging. VBUS reaches the +5V node through Schottky diode D1 (SDM2U30CSP-7B, 2 A).
- **VIN:** external supply input that reaches the +5V node through Schottky diode D4 (SDM2U30CSP-7B, 2 A).

- **+5V:** input of the BQ24074 charger and power-path controller.
- **VBAT:** battery domain intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V.
- **BMS\_OUT:** BQ24074 power-path output used as the system supply and as the input of the 3.3 V buck regulator.
- **3.3 V:** regulated logic supply generated by the SGM6029. This rail powers the ESP32-C5HR8, external flash, sensors, fuel gauge, microSD interface, QWIIC and J20 connectors, RGB LED, and indicators.

The SGM6029 is rated for up to 1 A output current. This current is shared between the ESP32-C5HR8, onboard peripherals, LEDs, and any external device powered from the 3.3 V rail. Radio transmission produces short current peaks, so the current available to external loads depends on the active board configuration and application workload.

The BQ24074RGTR manages single-cell LiPo battery charging. The schematic annotates a nominal charge current of 250 mA (ICHARG=250mA), set by R13. The input current limit is set by R12. Both values may be changed by modifying the corresponding resistor according to the BQ24074 configuration requirements.

All onboard digital interfaces operate in the 3.3 V logic domain. QWIIC, J20, microSD, and exposed GPIO signals must therefore be used with 3.3 V-compatible devices unless external level translation is provided.

Connector mechanical current ratings must not be interpreted as board power ratings. Maximum usable current is also constrained by the associated PCB traces, power path, regulator, protection components, and operating conditions.

## 2.3 Digital Interfaces

All onboard digital peripherals are connected to the board's 3.3 V logic domain. External logic compatibility must nevertheless be checked against the ESP32-C5 input/output specifications and the power state of both devices.

| Interface       | Board connection              | Electrical note                                                 |
|-----------------|-------------------------------|-----------------------------------------------------------------|
| Sensor I2C      | GPIO2 / GPIO3                 | Onboard sensors and fuel gauge; 4.7 kΩ pull-ups; strapping pins |
| QWIIC I2C       | GPIO1 / GPIO6                 | Connector also supplies 3.3 V and GND                           |
| SPI / microSD   | GPIO27, GPIO7, GPIO26, GPIO23 | SCK, MOSI, MISO, microSD CS; shared with J20                    |
| J20 chip select | GPIO10                        | SS for an external SPI device                                   |
| WS2812 data     | GPIO0                         | Drives the onboard RGB LED                                      |
| UART0           | GPIO11 / GPIO12               | TX through 499 Ω series resistor                                |
| USB             | GPIO13 / GPIO14               | Native USB Serial/JTAG                                          |

## 2.4 Confirmed Component Values

| Symbol / parameter | Value           | Scope                 |
|--------------------|-----------------|-----------------------|
| ESP32-C5 HP SRAM   | 384 KB          | ESP32-C5HR8 component |
| ESP32-C5 CPU clock | 240 MHz maximum | ESP32-C5HR8 component |

| Symbol / parameter  | Value   | Scope                 |
|---------------------|---------|-----------------------|
| In-package PSRAM    | 8 MB    | ESP32-C5HR8 component |
| QSPI flash capacity | 64 Mbit | IC3                   |
| Buck output current | 1 A     | U2 component rating   |
| Charge current      | 250 mA  | R13 = 3.57 kΩ         |
| Crystal frequency   | 48 MHz  | XTAL1                 |
| Sensor I2C pull-ups | 4.7 kΩ  | R15, R16              |

## 2.5 Application-Dependent and Uncharacterized Parameters

Some electrical and performance characteristics depend on firmware configuration, operating mode, connected peripherals, and application load. Unless otherwise specified, these parameters should not be interpreted as guaranteed board-level limits.

- **Available 3.3 V current for external loads:** the SGM6029 provides up to 1 A total output current. This current is shared by the ESP32-C5HR8, radio transmission peaks, flash, sensors, LEDs, and external 3.3 V loads.
- **Board current consumption:** total current consumption varies with CPU activity, radio mode and transmit power, enabled peripherals, RGB LED brightness, microSD operation, and connected external devices.
- **RF performance:** radiated performance depends on the external antenna, cable, enclosure, and the populated matching network. It is not characterized at board level.
- **Low-battery operation:** the SGM6029 input is the BQ24074 system output. 3.3 V regulation at low battery voltage is not characterized at board level.
- **GPIO electrical characteristics:** GPIO drive strength, input thresholds, and other pin-level electrical characteristics follow the ESP32-C5 specifications unless a board-level circuit modifies the corresponding signal.
- **Environmental characteristics:** board-level operating temperature, humidity, ESD immunity, and thermal performance are not specified unless explicitly stated in a released qualification or validation specification.

## 2.6 Electrical Precautions

1. Use only the specified power inputs: 5 V nominal through USB-C or VIN, or a 3.7 V nominal / 4.2 V maximum single-cell LiPo battery.
2. All GPIO and onboard digital interfaces operate in the 3.3 V logic domain. **Do not apply 5 V directly to GPIO, QWIIIC, J20, microSD, or other 3.3 V signal lines.**
3. Observe the battery connector polarity before connecting a single-cell LiPo battery. Do not connect multi-cell battery packs or batteries exceeding 4.2 V. Do not connect batteries to JP1 and J3 at the same time; both connectors are wired in parallel.
4. Do not externally drive the 3.3 V rail while the onboard regulator is active unless the external power source and operating configuration have been specifically designed for this purpose.
5. Connect external circuits to a common ground before applying external clock, data, or control signals.
6. Do not hold GPIO2, GPIO3, GPIO7, GPIO25, GPIO26, GPIO27, or GPIO28 at a level that changes the boot configuration during reset; these are ESP32-C5 strapping pins.

7. Connect an antenna to J2 before enabling radio transmission.
8. Ensure that the total load on the 3.3 V rail, including onboard circuitry and externally connected devices, remains within the 1 A capability of the SGM6029 regulator.

## 3 Functional Overview

The UNIT PULSAR ESP32-C5 is organized around the ESP32-C5HR8 SoC, with an external flash, a dual-band RF path, fixed onboard sensors, and a battery power path. Firmware can use each block independently or combine wireless connectivity, sensing, storage, and user feedback in one application.

### 3.1 Block Diagram

A block-diagram drawing for revision V1.2.3 is not yet available. The functional relationships are:

```

USB-C / VIN / battery
|
BQ24074 power path ---- MAX17048 fuel gauge
|
SGM6029 buck -> 3.3 V
|
ESP32-C5HR8 (8 MB PSRAM in package)
+-----+-----+-----+-----+-----+
|         |         |         |         |         |
| QSPI    | RF      | I2C     | SPI     | WS2812  | GPIO
| flash   | diplexer| sensors | microSD| RGB      | + QWIIC
|         | + J2    |         | + J20   |         |

```

The ESP32-C5HR8 controls every data path. The BY25Q64 flash stores executable code and nonvolatile application data. The in-package PSRAM extends working memory. The microSD socket provides removable storage. The pressure, humidity/temperature, proximity/light, and magnetic sensors and the battery fuel gauge share an internal I2C bus. QWIIC, J20, and the edge pads expose expansion signals, while the RF connector and RGB LED provide wireless and visual outputs.

### 3.2 Board Topology

| RefDes | Component         | Confirmed role                                  |
|--------|-------------------|-------------------------------------------------|
| IC1    | ESP32-C5HR8       | Main SoC with in-package 8 MB PSRAM             |
| IC3    | BY25Q64ESHIG      | 64 Mbit (8 MiB) QSPI flash memory               |
| IC2    | BQ24074RGTR       | Single-cell Li-Ion/LiPo charger with power path |
| U2     | SGM6029CYG        | 4 MHz, 1 A buck regulator for the 3.3 V rail    |
| L2     | 470 nH inductor   | SGM6029 output inductor                         |
| IC4    | MAX17048G+T10     | Battery fuel gauge                              |
| A2     | LPS22HBTR         | Barometric pressure sensor                      |
| U\$36  | FHT40-DD-TR       | Humidity and temperature sensor                 |
| U3     | VCNL4040M3OE      | Proximity and ambient-light sensor              |
| U4     | MMC5603NJ         | Three-axis magnetometer                         |
| U1     | LFD182G45DCHD277  | 2.4 GHz / 5 GHz RF diplexer                     |
| J2     | RF1-2626-125CE-CT | RF coaxial antenna connector                    |

| RefDes          | Component          | Confirmed role                                 |
|-----------------|--------------------|------------------------------------------------|
| XTAL1           | 48 MHz crystal     | ESP32-C5 main clock source                     |
| MICRO_SD-HOLDER | 47309-2651         | microSD card socket on SPI signals             |
| LED1            | WS2812 1010        | Addressable RGB LED                            |
| J1              | HCZZ0032-4         | Four-position, 1 mm pitch QWIIIC connector     |
| J20             | SM06B-SRSS-TB      | Six-position, 1 mm pitch SPI connector         |
| NANO1           | USB-C connector    | USB power, programming, and USB data interface |
| JP1, J3         | Battery connectors | PH 2.0 mm and 1.25 mm, wired in parallel       |
| S1              | BOOT switch        | Pulls GPIO28 low to enter download boot mode   |
| S2              | RESET switch       | Pulls CHIP_PU low                              |
| D1, D4          | SDM2U30CSP-7B      | Schottky diodes from VBUS and VIN to +5V       |

The board also provides solder jumpers that connect optional signals:

| Solder jumper function | Connection                                      |
|------------------------|-------------------------------------------------|
| 3.3 V regulator enable | SGM6029 EN to header position 3 (EN)            |
| Battery access         | VBAT to header position 4                       |
| RGB chain extension    | LED1 D0 to header position 11                   |
| Ground                 | GND to header position 13 (Nano RESET position) |
| Sensor bus             | SDA_SENSOR / SCL_SENSOR to GPIO2 / GPIO3        |
| Fuel-gauge alert       | MAX17048 ALERT to D5 / GPIO23                   |
| Fuel-gauge quick start | MAX17048 QSTRT to D6 / GPIO24                   |
| Charger power good     | BQ24074 PG00D to D7 / GPIO25                    |
| Proximity interrupt    | VCNL4040 INT to A6 / GPIO6                      |

The default state of each solder jumper is not stated in the schematic; verify it on the assembled board.

The two edge-pad rows expose power, analog, serial, SPI, and general-purpose signals. Several of these signals are shared with onboard devices, so firmware ownership must be decided before reusing them.

### 3.3 Processor

The ESP32-C5HR8 is the main controller. According to the Espressif component datasheet, the device provides a high-performance 32-bit RISC-V core up to 240 MHz, a low-power RISC-V core up to 48 MHz, 384 KB of HP SRAM, 16 KB of LP SRAM, and 320 KB of ROM.

The radio supports dual-band 2.4 GHz and 5 GHz Wi-Fi 6 (802.11ax) together with 802.11a/b/g/n, Bluetooth LE (Bluetooth Core 6.0 certified), and IEEE 802.15.4 for Thread 1.4 and Zigbee 3.0.

The peripheral set includes a 12-bit SAR ADC with up to six channels, UART, I2C, I2S, a general-purpose SPI port, USB Serial/JTAG, two CAN FD controllers, LED PWM, MCPWM, RMT, PARLIO, and GDMA. The PULSAR design routes a selected subset of these functions to onboard devices and external connectors.

The ESP32-C5 does not include an SD host controller; the microSD socket is therefore wired for SPI mode.

### 3.4 Memory Architecture: Flash and PSRAM

The BY25Q64ESHIG provides 64 Mbit (8 MiB) of external quad SPI flash on the dedicated flash pins (GPIO15–GPIO18 and GPIO20–GPIO22). It stores the bootloader, partition table, application firmware, and any filesystem or data region defined by the selected partition scheme. These pins are not available for other uses.

The ESP32-C5HR8 includes 8 MB of quad SPI PSRAM inside the package. In the Arduino workflow, PSRAM must be enabled in the board options; ESP-IDF enables it through `menuconfig`. PSRAM is suited to network buffers, file buffers, sensor history, and large application structures. Interrupt state and timing-critical data should remain in internal SRAM. The application must check allocation results.

### 3.5 RF Path and Antenna Connector

The ESP32-C5HR8 provides separate RF ports for the 2.4 GHz band (ANT\_2G, pin 42) and the 5 GHz band (ANT\_5G, pin 48). The U1 LFD182G45DCHD277 diplexer combines both ports into a single path covering 2.4–2.5 GHz and 4.9–5.95 GHz, which is routed to the J2 RF1-2626-125CE-CT coaxial connector.

| Path                  | Series element   | Shunt positions          |
|-----------------------|------------------|--------------------------|
| J2 to diplexer common | L5 (0 $\Omega$ ) | C20, C21 (not populated) |
| Diplexer LF to ANT_2G | L1 (0 $\Omega$ ) | C7, C9 (not populated)   |
| Diplexer HF to ANT_5G | L3 (0 $\Omega$ ) | C18, C4 (not populated)  |

The 0  $\Omega$  links and unpopulated capacitor positions form tuning networks that can be adjusted for a specific antenna. No onboard antenna is fitted. Connect a dual-band antenna to J2 before enabling radio transmission.

### 3.6 microSD Card Socket

The onboard 47309-2651 microSD socket is connected in SPI mode and operates in the 3.3 V logic domain.

| Socket signal | GPIO | SPI role      |
|---------------|------|---------------|
| CLK           | 27   | SCK           |
| CMD           | 7    | MOSI          |
| DAT0          | 26   | MISO          |
| DAT3          | 23   | Chip select   |
| DAT1, DAT2    | —    | Not connected |

SCK, MOSI, and MISO are shared with the J20 connector and the D11–D13 header positions. External SPI devices must use their own chip select, such as GPIO10 on J20. GPIO23 is also the optional MAX17048 ALERT input.

File writes should be flushed and closed before card extraction or power removal to reduce the risk of filesystem corruption. FAT32 is recommended.

### 3.7 LED Indicators

LED1 is a WS2812 1010 addressable RGB LED driven by GPIO0 (NEOP\_DIN), which is also available at header position 20. A solder jumper can connect LED1 D0 to header position 11, allowing the chain to be extended with compatible external addressable LEDs.

The board also includes a user indicator (BUILTIN2, pink) on GPIO27 through a 10 kΩ resistor, a red power indicator on the 3.3 V rail, and an orange charge-status indicator driven by the BQ24074 CHG output.

Applications should limit RGB brightness when power consumption or thermal rise is important.

### 3.8 BQ24074 and SGM6029 Power Management System

IC2 is a BQ24074RGTR single-cell Li-Ion/LiPo charger with dynamic power-path management. Its IN pin is fed from the +5V node, which combines USB VBUS (through D1) and VIN (through D4). Its OUT pin (BMS\_OUT) supplies the system while the battery charges.

| BQ24074 pin | Board configuration                          |
|-------------|----------------------------------------------|
| ISET        | R13 = 3.57 kΩ; 250 mA nominal charge current |
| ILIM        | R12 = 1.1 kΩ; ≈1.4 A input current limit     |
| EN1, EN2    | GND and BMS_OUT; input limit set by ILIM     |
| CE          | GND; charging enabled                        |
| TS          | 10 kΩ to GND (R11); no battery thermistor    |
| CHG         | Orange charge LED                            |
| PGOOD       | Optional solder jumper to D7 / GPIO25        |

U2 is an SGM6029CYG 4 MHz synchronous buck regulator. It converts BMS\_OUT into the regulated 3.3 V rail, with the output voltage selected by R14 on the VSEL /MODE pin. The regulator is rated for up to 1 A. Its EN pin is pulled up to BMS\_OUT by R8 (10 kΩ) and can be connected to header position 3 through a solder jumper for external control of the 3.3 V rail.

The battery input is intended for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. JP1 (PH 2.0 mm) and J3 (1.25 mm) are wired in parallel to VBAT.

### 3.9 Power Tree

```

USB-C VBUS --D1--+
      +--> +5V --> BQ24074 IN
VIN -----D4-----+
                      |
                      +--> BMS_OUT --> SGM6029 --> 3.3 V
VBAT (JP1 / J3) <--> BQ24074 BAT
      |
      +--> MAX17048 CELL / VDD
    
```

The power tree summarizes the main supply domains and their relationship. It is intended as a functional reference and does not replace the complete schematic.

The 3.3 V rail is available through the edge pads, the QWIIC connector, and J20. The total 3.3 V load, including radio transmission peaks, must remain within the 1 A regulator output capability.

### 3.10 Environmental and Magnetic Sensors

Four sensors share the internal SDA\_SENSOR / SCL\_SENSOR I2C bus with the fuel gauge. R15 and R16 (4.7 kΩ) pull the bus up to 3.3 V, and solder jumpers connect it to GPIO2 (SDA) and GPIO3 (SCL).

| Device              | Measurement                       | I2C address | Notes                                                   |
|---------------------|-----------------------------------|-------------|---------------------------------------------------------|
| LPS22HBTR (A2)      | Barometric pressure, 260–1260 hPa | 0x5C        | SA0 tied to GND; INT_DRDY not connected                 |
| FHT40-DD-TR (U\$36) | Relative humidity and temperature | 0x44        | SHT40-compatible default; verify for the assembled part |
| VCNL4040M3OE (U3)   | Proximity and ambient light       | 0x60        | INT to GPIO6 through optional solder jumper             |
| MMC5603NJ (U4)      | Three-axis magnetic field         | 0x30        | Compass and orientation applications                    |

Typical applications include weather stations, indoor air monitoring, altitude estimation, gesture or presence detection, automatic display brightness, and compass headings. Measurement range, filtering, output data rate, and interrupt behavior are configured through the sensor registers or software driver and are not fixed by the PCB.

### 3.11 MAX17048 Fuel Gauge

IC4 is a MAX17048G+T10 single-cell fuel gauge with ModelGauge. Its CELL and VDD pins connect to VBAT, and it communicates on the sensor I2C bus at address 0x36. Firmware can read battery voltage and state of charge without an external sense resistor.

The ALERT output can be connected to D5 / GPIO23 and the QSTRT input to D6 / GPIO24 through optional solder jumpers. Because GPIO23 is also the microSD chip select, applications using both functions must account for this shared connection before closing the ALERT jumper.

### 3.12 I2C and QWIIC Expansion

The UNIT PULSAR ESP32-C5 provides two independent I2C routes, allowing the onboard sensors and external QWIIC devices to operate on separate buses.

| Interface  | SDA   | SCL   | Primary use                                      |
|------------|-------|-------|--------------------------------------------------|
| Sensor I2C | GPIO2 | GPIO3 | Onboard sensors, fuel gauge, and edge-pad access |
| QWIIC I2C  | GPIO1 | GPIO6 | External QWIIC devices                           |

GPIO2 and GPIO3 are also exposed at the SDA / SCL edge pads and are ESP32-C5 strapping pins (MTMS, MTDI). GPIO1 and GPIO6 are also exposed at the A1 and A6 edge pads.

Both interfaces operate in the 3.3 V logic domain. Device addresses, pull-up configuration, bus capacitance, and cable length must be considered when multiple devices are connected to the same bus.

### 3.13 Combined System Operation

The UNIT PULSAR ESP32-C5 is designed to support concurrent operation of its onboard subsystems. The ESP32-C5HR8 can combine wireless connectivity, environmental sensing, battery monitoring, removable storage, and RGB indication within the same application.

For example, sensor data can be buffered in PSRAM, logged to the microSD card, and published over Wi-Fi or Thread, while the fuel gauge reports battery state and the RGB LED indicates connection status.

Applications using the radio together with storage and sensors should consider current peaks during transmission, memory allocation, task priorities, and storage write latency. Shared GPIO functions must also be considered when enabling optional solder-jumper connections.

## 4 Connectors & Pinouts

Pin labels describe hardware revision V1.2.3. GPIO numbers identify ESP32-C5HR8 signals according to the Espressif datasheet pin table; framework aliases depend on the selected board definition.

### 4.1 General Pinout

| Pos. | Board label     | ESP32-C5HR8      | Primary board role         | Shared resource                               |
|------|-----------------|------------------|----------------------------|-----------------------------------------------|
| 1    | D13 / SCK / LED | GPIO27           | SPI clock                  | microSD CLK, J20, BUILTIN2 LED; strapping pin |
| 2    | 3V3             | —                | 3.3 V rail                 | SGM6029 output                                |
| 3    | EN              | —                | 3.3 V regulator enable     | Solder jumper to SGM6029 EN                   |
| 4    | VBAT            | —                | Battery rail               | Solder jumper to VBAT                         |
| 5    | A1 / D15        | GPIO1 / ADC1_CH0 | Analog-capable I/O         | QWIIC SDA                                     |
| 6    | A2 / D16        | GPIO4 / ADC1_CH3 | Analog-capable I/O         | MTCK                                          |
| 7    | A3 / D17        | GPIO5 / ADC1_CH4 | Analog-capable I/O         | MTDO                                          |
| 8    | SDA / D18       | GPIO2 / ADC1_CH1 | I2C data                   | Sensor bus; MTMS strapping pin                |
| 9    | SCL / D19       | GPIO3 / ADC1_CH2 | I2C clock                  | Sensor bus; MTDI strapping pin                |
| 10   | A6 / D21        | GPIO6 / ADC1_CH5 | Analog-capable I/O         | QWIIC SCL; optional VCNL4040 INT              |
| 11   | NEOPIXEL DOUT   | —                | RGB-chain extension output | Solder jumper to LED1 D0                      |
| 12   | VBUS            | —                | USB VBUS                   | USB-C supply                                  |
| 13   | GND             | —                | Ground                     | Solder jumper (Nano RESET position)           |
| 14   | GND             | —                | Ground                     | Common return                                 |
| 15   | VIN             | —                | Supply input               | To +5V through D4                             |
| 16   | TX0 / D1        | GPIO11           | UART0 TX                   | 499 $\Omega$ series resistor R3               |
| 17   | RX0 / D0        | GPIO12           | UART0 RX                   | General-purpose                               |
| 18   | RST             | CHIP_PU          | Reset                      | RESET button, 10 k $\Omega$ pull-up           |
| 19   | GND             | —                | Ground                     | Common return                                 |
| 20   | D2 / RGB        | GPIO0            | NeoPixel data              | LED1 DI                                       |
| 21   | D3 / BOOT       | GPIO28           | BOOT input                 | BOOT button; strapping pin                    |
| 22   | D4              | GPIO9            | Digital I/O                | General-purpose                               |

| Pos. | Board label | ESP32-C5HR8 | Primary board role | Shared resource                                   |
|------|-------------|-------------|--------------------|---------------------------------------------------|
| 23   | D5          | GPIO23      | Digital I/O        | microSD CS; optional MAX17048 ALERT               |
| 24   | D6          | GPIO24      | Digital I/O        | Optional MAX17048 QSTRT                           |
| 25   | D7          | GPIO25      | Digital I/O        | Optional BQ24074 PG00D; strapping pin             |
| 26   | D8          | USB D-      | USB D-             | Shared with USB-C; GPIO13 through 22 $\Omega$ R9  |
| 27   | D9          | USB D+      | USB D+             | Shared with USB-C; GPIO14 through 22 $\Omega$ R10 |
| 28   | D10 / SS    | GPIO10      | SPI chip select    | J20 SS                                            |
| 29   | D11 / MOSI  | GPIO7       | SPI MOSI           | microSD CMD, J20; strapping pin                   |
| 30   | D12 / MISO  | GPIO26      | SPI MISO           | microSD DAT0, J20; strapping pin                  |

EN, VBAT, NEOPIXEL DOUT, and the ground at position 13 are connected to their edge-pad positions through solder jumpers.

## 4.2 Interface Summary

| Interface               | Signal  | Board label | GPIO   | Notes                                                         |
|-------------------------|---------|-------------|--------|---------------------------------------------------------------|
| UART0                   | RX      | D0          | GPIO12 |                                                               |
| UART0                   | TX      | D1          | GPIO11 | 499 $\Omega$ series resistor R3                               |
| SPI                     | SS / CS | D10         | GPIO10 | Also J20 pin 4                                                |
| SPI                     | MOSI    | D11         | GPIO7  | Shared with microSD and J20                                   |
| SPI                     | MISO    | D12         | GPIO26 | Shared with microSD and J20                                   |
| SPI                     | SCK     | D13         | GPIO27 | Shared with microSD, J20, and BUILTIN2 LED                    |
| microSD                 | CS      | D5          | GPIO23 | Separate chip select from D10                                 |
| Sensor I2C (SDA_SENSOR) | SDA     | SDA / D18   | GPIO2  | Onboard sensors through solder jumper; 4.7 k $\Omega$ pull-up |
| Sensor I2C (SCL_SENSOR) | SCL     | SCL / D19   | GPIO3  | Onboard sensors through solder jumper; 4.7 k $\Omega$ pull-up |
| QWIIC I2C               | SDA     | A1 / D15    | GPIO1  | J1 pin 3; no onboard pull-up                                  |

| Interface | Signal | Board label | GPIO   | Notes                        |
|-----------|--------|-------------|--------|------------------------------|
| QWIIC I2C | SCL    | A6 / D21    | GPIO6  | J1 pin 4; no onboard pull-up |
| USB       | D-     | D8          | GPIO13 | Native USB; 22 $\Omega$ R9   |
| USB       | D+     | D9          | GPIO14 | Native USB; 22 $\Omega$ R10  |

The SDA / SCL edge pads are the sensor bus, not a separate external bus: any device connected there shares the bus with the five onboard I2C devices. QWIIC is an independent bus on GPIO1 and GPIO6.

Two schematic net labels differ from the connections drawn at the header:

- **D2:** IC1 GPIO8 carries the net label D2, but header position 20 (D2) is wired to NEOP\_DIN / GPIO0. No other connection is shown for the GPIO8 net; treat GPIO8 as unassigned until the PCB is verified.
- **A6:** the net is named A6/GPIO4/TOUCH4, but it connects to IC1 pin 15, which is GPIO6. GPIO4 is A2 / D16 (pin 13). The ESP32-C5 has no touch sensor, so TOUCH4 does not apply.

### 4.3 Analog Inputs

| Analog label | Digital label | GPIO  | ADC channel | Shared use                       |
|--------------|---------------|-------|-------------|----------------------------------|
| A1           | D15           | GPIO1 | ADC1_CH0    | QWIIC SDA                        |
| A2           | D16           | GPIO4 | ADC1_CH3    | —                                |
| A3           | D17           | GPIO5 | ADC1_CH4    | —                                |
| A4 / SDA     | D18           | GPIO2 | ADC1_CH1    | Sensor I2C SDA                   |
| A5 / SCL     | D19           | GPIO3 | ADC1_CH2    | Sensor I2C SCL                   |
| A6           | D21           | GPIO6 | ADC1_CH5    | QWIIC SCL; optional VCNL4040 INT |

The Nano A0 and A7 positions carry VBAT and NEOPixel DOUT and are not analog inputs. A4/A5 and A1/A6 are usable as analog inputs only while the corresponding I2C bus is not in use.

### 4.4 Board Resources and I2C Addresses

| Resource     | Signal / GPIO                | Bus             | I2C address | Notes                                    |
|--------------|------------------------------|-----------------|-------------|------------------------------------------|
| Native USB   | GPIO13 / GPIO14              | USB Serial/JTAG | —           | D8 / D9                                  |
| BOOT button  | GPIO28                       | —               | —           | D3; strapping pin                        |
| UART0        | GPIO11 / GPIO12              | UART            | —           | D1 / D0                                  |
| NeoPixel in  | GPIO0 (NEOP_DIN)             | —               | —           | LED1 WS2812 1010; D2                     |
| NeoPixel out | LED1 D0 (NEOP_D0)            | —               | —           | Header position 11 through solder jumper |
| Flash        | GPIO15–GPIO18, GPIO20–GPIO22 | SPI0/1          | —           | BY25Q64; not available as GPIO           |
| PSRAM        | In package                   | SPI0/1          | —           | 8 MB, ESP32-C5HR8                        |

| Resource          | Signal / GPIO                    | Bus        | I2C address | Notes                                                      |
|-------------------|----------------------------------|------------|-------------|------------------------------------------------------------|
| microSD           | GPIO27, GPIO7, GPIO26, GPIO23    | SPI        | —           | CS on D5                                                   |
| QWIIC             | GPIO1 / GPIO6                    | QWIIC I2C  | —           | J1                                                         |
| Battery monitor   | GPIO2 / GPIO3                    | Sensor I2C | 0x36        | MAX17048G+T10                                              |
| Light / proximity | GPIO2 / GPIO3                    | Sensor I2C | 0x60        | VCNL4040M3OE                                               |
| Temperature / RH  | GPIO2 / GPIO3                    | Sensor I2C | 0x44        | FHT40-DD-TR (SHT40 in schematic); verify address for FHT40 |
| Pressure          | GPIO2 / GPIO3                    | Sensor I2C | 0x5C        | LPS22HBTR (LPS22DFTR in schematic); SA0 to GND             |
| Magnetometer      | GPIO2 / GPIO3                    | Sensor I2C | 0x30        | MMMC5603NJ                                                 |
| RF                | ANT_2G (pin 42), ANT_5G (pin 48) | —          | —           | U1 diplexer to J2                                          |

## 4.5 Arduino Nano Pinout Compatibility

The board uses two parallel 15-position edge rows based on the Arduino Nano layout. This provides a familiar mechanical footprint and labeling pattern, but does not imply full electrical or shield compatibility.

Important differences include:

- The ESP32-C5HR8 and board GPIO operate in the 3.3 V logic domain.
- Several conventional positions have PULSAR-specific functions: EN at AREF, VBAT at A0, NEOPIXEL DOUT at A7, VBUS at 5V, and a jumper-selected ground at the second RESET position.
- D2 carries the RGB LED data and D3 carries the BOOT signal.
- D8 and D9 carry the ESP32-C5 native USB D-/D+ signals and should not be treated as conventional digital I/O.
- GPIO2, GPIO3, GPIO7, GPIO25, GPIO26, GPIO27, and GPIO28 are strapping pins. External circuits must not force a level that changes the boot mode during reset.
- Analog channel labels do not follow a sequential GPIO order.

Before installing the board into a Nano-style carrier, verify power, reset, analog, digital, and mechanically shared positions.

## 4.6 QWIIC Connector

J1 is a four-position, 1 mm-pitch QWIIC connector carrying:

| Pin | Signal | ESP32-C5HR8 connection | Function           |
|-----|--------|------------------------|--------------------|
| 1   | GND    | Ground                 | Common return      |
| 2   | 3.3 V  | Regulated rail         | Peripheral supply  |
| 3   | SDA    | GPIO1                  | External I2C data  |
| 4   | SCL    | GPIO6                  | External I2C clock |

The QWIIC interface operates in the 3.3 V logic domain and is separate from the GPIO2/GPIO3 sensor bus. The board does not fit pull-up resistors on GPIO1/GPIO6; QWIIC devices normally provide them.

The QWIIC 3.3 V supply is provided by the SGM6029 regulator, which supports up to 1 A total output current shared between the onboard circuitry and all external 3.3 V loads.

## 4.7 J20 SPI Connector

J20 is a six-position, 1 mm-pitch SH connector for external SPI devices.

| Pin | Signal | ESP32-C5HR8 connection |
|-----|--------|------------------------|
| 1   | SCK    | GPIO27                 |
| 2   | MOSI   | GPIO7                  |
| 3   | MISO   | GPIO26                 |
| 4   | SS     | GPIO10                 |
| 5   | 3.3 V  | Regulated rail         |
| 6   | GND    | Ground                 |

SCK, MOSI, and MISO are shared with the microSD socket. Keep the microSD chip select (GPIO23) high while communicating with a J20 device.

## 4.8 MicroSD Connector

| Socket signal | GPIO   | Description                            |
|---------------|--------|----------------------------------------|
| CLK           | GPIO27 | SPI SCK                                |
| CMD           | GPIO7  | SPI MOSI                               |
| DAT0          | GPIO26 | SPI MISO                               |
| DAT3          | GPIO23 | SPI chip select                        |
| DAT1, DAT2    | —      | Not connected                          |
| VDD           | 3.3 V  | Card supply                            |
| GND / shield  | Ground | Electrical return and connector shield |

Insert or remove the microSD card only when filesystem activity has stopped. Software should flush and close open files before card extraction or power removal.

## 4.9 Battery Connections

JP1 (PH 2.0 mm) and J3 (1.25 mm) provide two battery connector options wired in parallel to VBAT. Connect only one battery at a time.

The battery input is designed for a single-cell LiPo battery with a nominal voltage of 3.7 V and a maximum fully charged voltage of 4.2 V. Charging is managed by the onboard BQ24074RGTR with a nominal charge current of 250 mA, and battery state is reported by the MAX17048 fuel gauge.

Always verify connector polarity before connecting a battery, as connector compatibility alone does not guarantee correct polarity.

A solder jumper can connect the battery rail to edge-pad position 4 (VBAT).

Do not connect multi-cell battery packs or batteries exceeding 4.2 V.

## 4.10 RF Antenna Connector

J2 is an RF1-2626-125CE-CT coaxial connector rated to 6 GHz. It carries the combined 2.4 GHz and 5 GHz signal from the U1 diplexer. Use a dual-band antenna with a mating plug and connect it before enabling radio transmission.

## 4.11 USB-C, BOOT, and Reset

The USB-C connector provides power, programming, and USB data connectivity. The USB D- and D+ signals connect to the ESP32-C5 USB Serial/JTAG controller (GPIO13 and GPIO14) through 22  $\Omega$  series resistors and are also exposed at the D8 and D9 positions. CC1 and CC2 have 5.1 k $\Omega$  pull-down resistors.

The BOOT button pulls GPIO28 low. Holding BOOT during reset selects the ESP32-C5 download boot mode used for firmware programming and recovery. The RESET button pulls CHIP\_PU low and restarts the ESP32-C5 while board power remains applied.

A typical firmware-recovery sequence is to hold BOOT, press and release RESET, then release BOOT after the device enumerates in download mode.

## 4.12 Debug Interface

The ESP32-C5 USB Serial/JTAG controller provides serial console, firmware download, and JTAG debugging through the USB-C connector without an external probe. The JTAG pins (MTMS, MTDI, MTCK, MTDO) are used as GPIO2–GPIO5 on this board and are not routed to dedicated debug pads.

# 5 Board Operation

The procedures below define the board workflow for the Arduino IDE and the Espressif Arduino core.

## 5.1 Getting Started with Arduino IDE

1. Install Arduino IDE 2.x.
2. Add the Espressif package index to **Additional Boards Manager URLs**: [https://espressif.github.io/arduino-esp32/package\\_esp32\\_index.json](https://espressif.github.io/arduino-esp32/package_esp32_index.json)
3. Install the **esp32 by Espressif Systems** package, version 3.3 or later (ESP32-C5 support).
4. Select **ESP32C5 Dev Module** or an installed PULSAR-specific board definition.
5. Set **Flash Size** to 8 MB, enable **PSRAM**, and enable **USB CDC On Boot** to use the USB-C serial port.
6. Connect an antenna to J2 if the sketch uses the radio.
7. Connect a USB-C data cable and select the enumerated serial port.
8. Open a basic Blink sketch and upload it.

Compile success confirms API and library compatibility; electrical operation also depends on the connected hardware and selected configuration.

## 5.2 First Power-up and Inspection

Use USB-C for initial power. Inspect component orientation, connector seating, and solder bridges before energizing the board. If available, use a current-limited source and monitor the 3.3 V rail.

Do not attach a battery, external VIN, microSD card, QWIIC peripheral, or J20 device for the first power check. Confirm the power indicator and absence of unexpected heating, then establish USB enumeration and upload behavior.

## 5.3 BOOT and Reset Workflow

For a normal update, the Arduino core resets the board into download mode automatically through USB Serial/JTAG. If automatic upload is unavailable:

1. Disconnect external driven signals.
2. Hold BOOT.
3. Press and release RESET, or connect USB while BOOT is held.
4. Release BOOT after the download port appears.
5. Upload the firmware using the toolchain.
6. Press RESET to run the application from external flash.

## 5.4 Recommended Bring-up Sequence

| Order | Subsystem | What the step establishes               |
|-------|-----------|-----------------------------------------|
| 1     | User LED  | Core execution, GPIO27, and upload path |

| Order | Subsystem              | What the step establishes                         |
|-------|------------------------|---------------------------------------------------|
| 2     | USB serial             | Host communication and diagnostic output          |
| 3     | ADC                    | Basic analog input path                           |
| 4     | RGB LED                | GPIO0 timing                                      |
| 5     | Sensor I2C             | GPIO2/GPIO3, solder jumpers, and sensor addresses |
| 6     | Fuel gauge             | MAX17048 voltage and state of charge              |
| 7     | QWIIC I2C              | GPIO1/GPIO6 and external expansion                |
| 8     | PSRAM                  | Capacity detection and allocation                 |
| 9     | microSD                | SPI bus, GPIO23 chip select, and filesystem       |
| 10    | Wi-Fi / BLE / 802.15.4 | Antenna, RF path, and network connection          |
| 11    | Combined application   | Memory ownership and subsystem coexistence        |

Testing one subsystem at a time makes pin conflicts and library configuration errors easier to isolate.

## 5.5 C++ Example Collection

C++ examples for the ESP32-C5 revision are planned under `software/cpp_examples/`. The recommended bring-up sequence in Section 5.4 defines the intended example coverage.

## 5.6 Memory Planning

Keep stacks, interrupt state, frequently accessed variables, and DMA descriptors in internal SRAM. Place large network buffers, logging queues, and historical data in PSRAM. Check every allocation and track free internal and PSRAM heap separately.

Wi-Fi, Bluetooth LE, and 802.15.4 stacks use internal memory; avoid allocating every maximum-size buffer at startup without a documented memory budget.

## 5.7 Storage Operation

Configure SPI on GPIO27 (SCK), GPIO7 (MOSI), and GPIO26 (MISO) with GPIO23 as the microSD chip select. Initialize the card, verify filesystem information, open only the files needed, call `flush()` after important writes, and close handles before removal or reset.

For data logging, buffer short records in RAM and write them in bounded batches rather than performing filesystem operations from an interrupt handler.

## 5.8 Radio, Sensor, and Storage Coexistence

The sensor bus (GPIO2/GPIO3), QWIIC bus (GPIO1/GPIO6), and SPI bus (GPIO7/GPIO26/GPIO27) do not overlap. The microSD card and J20 devices share the SPI bus and must use separate chip selects. All subsystems compete for processor time, memory, and power; radio transmission adds current peaks to the 3.3 V rail.

Use bounded update rates, buffered logging, and nonblocking network tasks. A combined application should expose health information over USB serial or RGB status rather than failing silently.

## 5.9 Troubleshooting Guide

| Symptom                  | Checks                                                         |
|--------------------------|----------------------------------------------------------------|
| No USB device            | Data-capable cable, BOOT/RESET sequence, host port, 3.3 V rail |
| Board does not boot      | Strapping pins held low by external circuits, BOOT button      |
| Blink does not run       | GPIO27 mapping, selected board, successful upload              |
| PSRAM unavailable        | PSRAM option enabled, allocation result                        |
| Sensors not found        | Sensor-bus solder jumpers, GPIO2/GPIO3, addresses              |
| microSD fails            | FAT32, GPIO23 chip select, SPI pins, card seating              |
| Weak or no wireless link | Antenna on J2, band selection, matching network                |
| Unstable combined app    | Internal heap, PSRAM use, blocking loops, pin conflicts        |

Record the exact board revision, core version, library versions, compiler output, power source, and reproduction steps for every bring-up issue.

## 6 Mechanical Information

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Hardware revision V1.2.3 uses a narrow development-board form factor with two parallel 15-position edge rows following the Arduino Nano template, USB-C at one end, and QWIIC, J20, and RF connectors for expansion.

### 6.1 Component-side Envelope

Board photographs for revision V1.2.3 are not yet available.

Adequate vertical and lateral clearance should be provided around the USB-C connector, pushbuttons, QWIIC and J20 connectors, RF connector, and other components. The edge pads must remain accessible when the board is mounted on headers, carrier boards, sockets, or directly soldered to another PCB.

### 6.2 Bottom-side Envelope

Adequate clearance must be maintained for battery connection, microSD card insertion and removal, and antenna cable routing. Enclosures and carrier boards must not cover the pressure, humidity, and light-sensor openings when those measurements are used.

### 6.3 Board Dimensions

A controlled dimension drawing and mounting-hole coordinates for revision V1.2.3 are not yet available. Do not derive dimensions from rendered board images.

### 6.4 Mechanical Integration Considerations

When integrating the UNIT PULSAR ESP32-C5 into a carrier board, fixture, or enclosure, provide sufficient clearance for:

- USB-C cable insertion and removal.
- QWIIC and J20 cable connection.
- Antenna cable and connector on J2.
- Battery connector and cable routing.
- microSD card insertion and removal.
- BOOT and RESET button access.
- Edge-pad soldering or header installation.
- Configuration solder-jumper access when required.
- Airflow and light paths to the environmental and optical sensors.

Keep metal enclosures, ground planes, and batteries away from the antenna. PCB thickness, maximum component heights, and connector insertion envelopes should be verified from the corresponding component specifications when they are critical to the final enclosure or carrier design.

## 7 Company Information

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| Item    | Information                                              |
|---------|----------------------------------------------------------|
| Company | UNIT Electronics                                         |
| Website | <a href="#">UNIT Electronics</a>                         |
| Address | Salvador 19, Cuauhtémoc, 06000 Mexico City, CDMX, Mexico |

The repository and Product Wiki in the next chapter provide the product's hardware, software, and reference documentation.

## 8 Reference Documentation

The following resources provide the official hardware, software, and application documentation for the UNIT PULSAR ESP32-C5.

| Reference                    | Location                                                 |
|------------------------------|----------------------------------------------------------|
| Product repository           | <a href="#">UNIT-Electronics-MX/unit_pulsar_esp32_c5</a> |
| Hardware reference           | <a href="#">hardware/README.md</a>                       |
| Technical wiki               | <a href="#">PULSAR ESP32-C5 Wiki</a>                     |
| Board schematic              | <a href="#">Hardware V1.2.3 schematic</a>                |
| ESP32-C5 component datasheet | <a href="#">Espressif ESP32-C5 Series Datasheet</a>      |
| Arduino core                 | <a href="#">Espressif Arduino-ESP32</a>                  |

The UNIT Electronics Wiki provides application-oriented documentation, setup procedures, peripheral usage examples, and software workflows for the board.

The hardware schematic and this Product Reference define the board-level implementation, connections, power architecture, interfaces, and hardware characteristics. ESP32-C5HR8 device-level electrical characteristics and processor specifications should be referenced from the official Espressif ESP32-C5 datasheet.

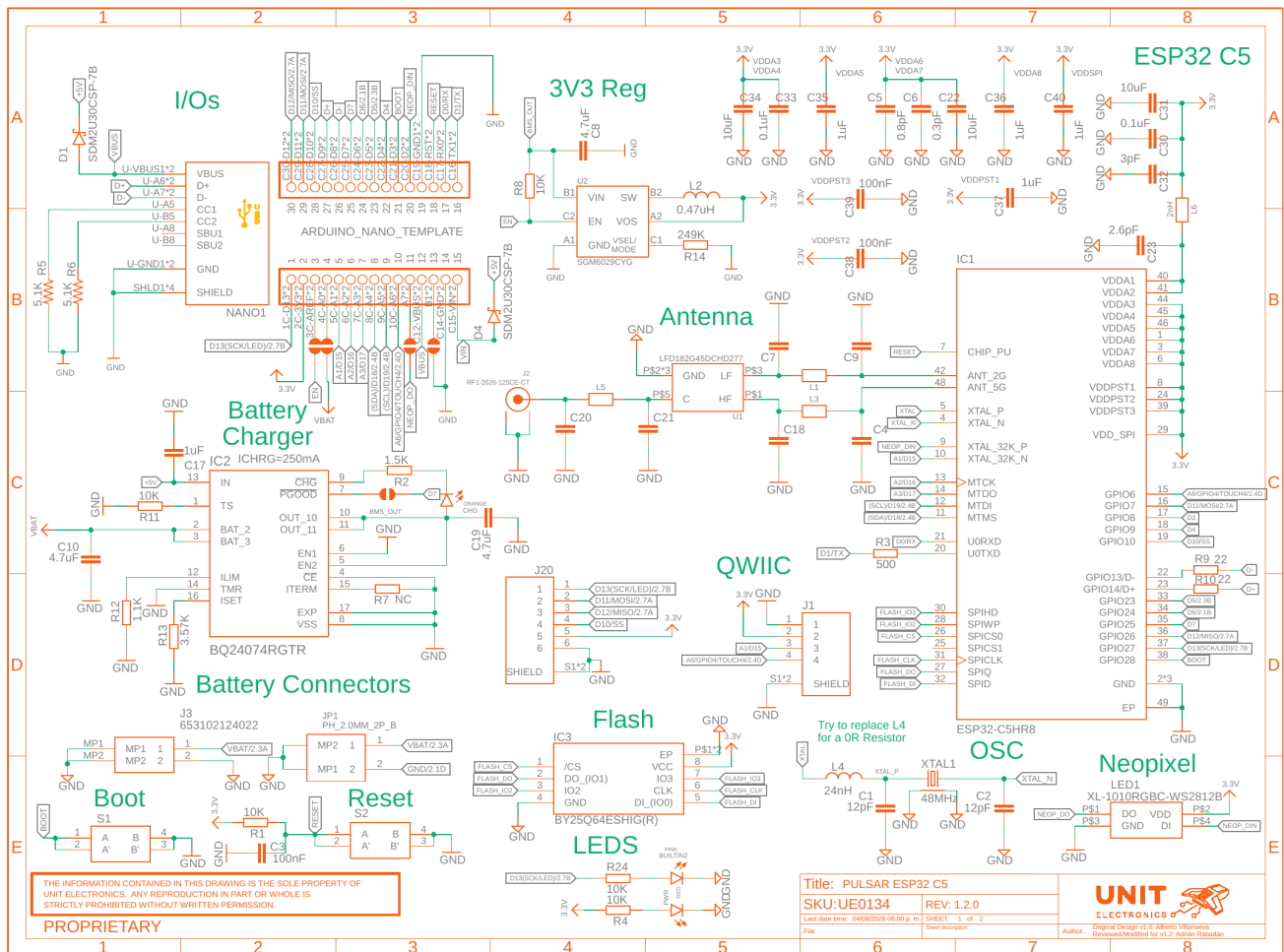
## 9 Appendix

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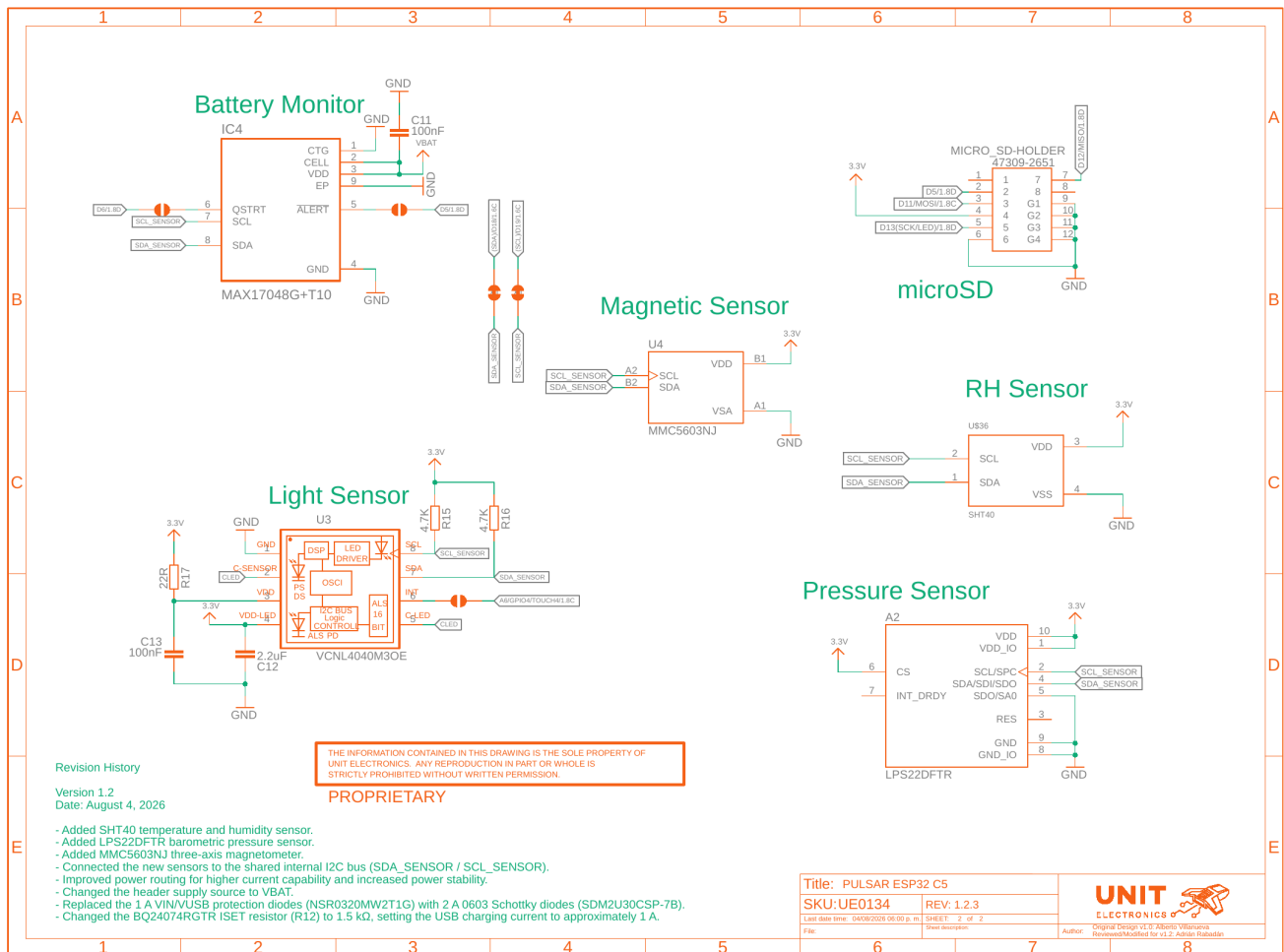
### 9.1 Schematic

The following two sheets are rendered from `unit_sch_v_1_2_3_ue0134_pulsar_esp32_c5.pdf`. The original PDF remains the authoritative electrical schematic for hardware revision V1.2.3.

## 9.1.1 I/Os, USB-C, Charger, 3.3 V Regulator, Antenna, ESP32-C5, Flash, and LEDs



## 9.1.2 Battery Monitor, microSD, and Sensors



## 9.2 Technical References

The documentation set for the UNIT PULSAR ESP32-C5 is organized around the following technical sources:

1. The hardware V1.2.3 schematic defines board connectivity, component designators, power architecture, and electrical implementation.
2. The bill of materials defines the assembled part numbers.
3. The official Espressif ESP32-C5 datasheet defines device-level electrical characteristics, processor capabilities, GPIO behavior, strapping pins, ADC operation, and radio functionality.
4. The UNIT Electronics Wiki provides setup procedures, peripheral usage documentation, application examples, and software workflows.

## 9.3 Design and Application Notes

The following characteristics depend on the final application, firmware configuration, connected peripherals, or mechanical integration:

- Total board current consumption varies with CPU activity, radio mode and transmit power, enabled peripherals, RGB LED brightness, microSD activity, and external loads.
- Available current for external 3.3 V devices depends on the portion of the SGM6029 1 A output capability consumed by the board itself.
- Radiated RF performance depends on the antenna, cable, enclosure, and the populated matching network.
- Environmental-sensor accuracy depends on self-heating, airflow, and enclosure design.
- Thermal performance and maximum component temperatures depend on workload, ambient conditions, airflow, enclosure design, and external loading.

## 9.4 Document Control

| Field             | Value                 |
|-------------------|-----------------------|
| Product           | UNIT PULSAR ESP32-C5  |
| SKU               | UE0134                |
| Product family    | UNIT DevLab ecosystem |
| Hardware revision | V1.2.3                |
| Product Reference | Version 0.1.0         |
| Publication date  | 2026-09-24            |

## 9.5 Source Notes

- The schematic title block shows REV: 1.2.0 on sheet 1 and REV: 1.2.3 on sheet 2; the schematic file name uses V1.2.3, and the revision history is labeled Version 1.2. This Product Reference uses V1.2.3.
- The schematic names the humidity sensor SHT40 and the pressure sensor LPS22DFTR; the bill of materials specifies FHT40-DD-TR and LPS22HBTR. This Product Reference follows the bill of materials for part numbers.
- The schematic revision history states that R12 was changed to 1.5 k $\Omega$  for an approximately 1 A USB charging current. The schematic and bill of materials show R12 = 1.1 k $\Omega$  on ILIM (input current limit) and R13 = 3.57 k $\Omega$  on ISET, with the annotation ICHRG=250mA. This Product Reference uses the schematic and bill-of-materials values.
- The J3 part number is 653102124022 in the schematic and 653102131822 in the bill of materials.
- GPIO8 carries the net label D2 at IC1, but no header connection is shown for that net. Header position D2 connects to GPIO0 (NEOP\_DIN).
- The net connected to header position 10 and QWIIC pin 4 is named A6/GPI04/TOUCH4, but it connects to IC1 pin 15 (GPIO6). The ESP32-C5 has no touch sensor. This Product Reference uses GPIO6.
- The label D21 for header position 10 is taken from the board pinout artwork; the schematic shows only A6.
- The ESP32-C5 does not provide an SD host controller; the microSD socket uses SPI mode and DAT1/DAT2 are not connected.

## 9.6 Hardware Errata

No hardware errata have been recorded for revision V1.2.3.